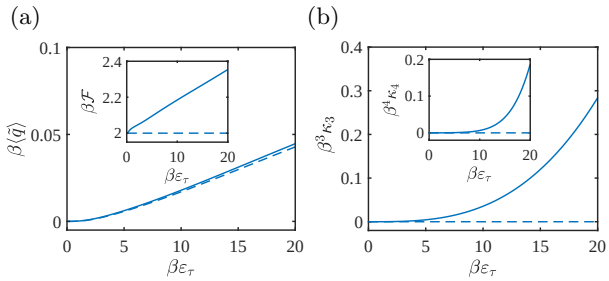

FIG. 2. Heat statistics of slow driving processes. (a) Mean excess heat (main) and Fano factor, $\mathcal{F}$ (inset). (b) Third cumulant (main) and fourth cumulant (inset) of the heat distribution, demonstrating non-Gaussian statistics. Solid lines show a quantum protocol with $\varepsilon_t = \varepsilon_0 + (\varepsilon_\tau - \varepsilon_0)\sin^2(\pi t/2\tau)$ and $\theta_t = \pi(t/\tau - 1)$, dashed lines show the corresponding classical protocol with identical ε_t but $\theta_t = 0$. The initial energy splitting is $\varepsilon_0 = 0.02\varepsilon_\tau$ and the protocol duration is $\tau = 250/\bar{\gamma}$, where $\bar{\gamma}$ is a characteristic thermalisation rate given by the time average of $\gamma_t = \frac{1}{2}\hbar^{-1}\alpha\varepsilon_t\coth(\beta\varepsilon_t/2)$, with $\alpha = 0.191$ the coupling to an Ohmic bath.

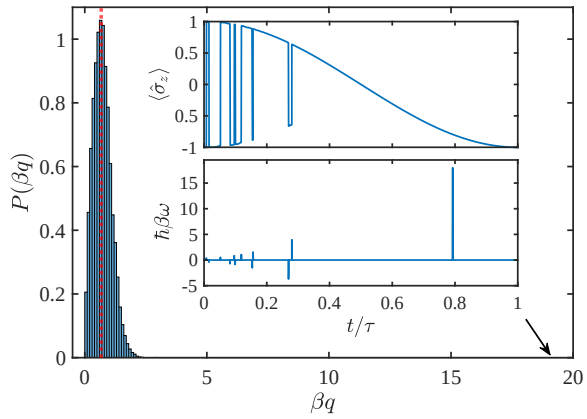

FIG. 3. Quantum-jump trajectory simulation of the coherent qubit erasure protocol of Fig. 2, with $\beta\varepsilon_\tau = 20$. Main panel: heat distribution over 3×10^4 trajectories, with the Landauer bound $\beta q = -\Delta S$ shown by the red dotted line. Inset: dynamics of a rare trajectory with large heat transfer, $q = 19.1k_BT$ (black arrow). Stochastic jumps in the otherwise continuous evolution of $\langle\hat{\sigma}_z\rangle$ (upper inset) are associated with the emission of energy quanta $\hbar\omega = \pm\varepsilon_t$ to the environment (lower inset). Non-adiabatic quantum evolution allows for two consecutive emissions and consequently extreme dissipation.

In Fig. 2 we plot the first four cumulants of the heat distribution, comparing a quantum protocol to a classical process with identical ε_t but $\dot{\theta}_t = 0$. These analytical results are derived in the slow-driving limit at order $\mathcal{O}(\tau^{-1})$ (see Appendix D). We show in Fig. 2(a) that the mean excess heat [Eq. (3)] takes small but non-zero values in the erasure regime, $\varepsilon_\tau \gg k_BT$, reflecting the entropy produced in this finite-time process. While the quantum and classical protocols show similar dissipation on average, they differ significantly in their fluctuations. The inset of Fig. 2 shows the Fano factor, $\mathcal{F} = \mathrm{Var}(\tilde{q})/\langle\tilde{q}\rangle$, which is increased by quantum fluctuations above the classical value, $\mathcal{F} = 2k_BT + \mathcal{O}(\tau^{-2})$, that follows from the fluctuation-dissipation relation. The most significant difference arises in higher-order statistics: the non-classical nature of the heat distribution is witnessed by its third and fourth cumulants, shown in Fig. 2(b). These imply significant skewness and kurtosis and signal the presence of non-Gaussian tails in the distribution.

To reveal the microscopic origin of these tails, we simulate individual runs of the erasure protocol using the quantum-jump trajectory approach (see Appendix E). A trajectory is described by a pure state $|\psi_t\rangle$ undergoing continuous time evolution interspersed by stochastic jumps, $|\psi_t\rangle \to |\pm\varepsilon_t\rangle$, where $\hat{H}_t\,|\pm\varepsilon_t\rangle = \pm\frac{1}{2}\varepsilon_t\,|\pm\varepsilon_t\rangle$. Each jump transfers a quantum of energy $\hbar\omega = \mp\varepsilon_t$ to the environment. The main panel of Fig. 3 shows the heat distribution obtained by numerically sampling many such trajectories for a quantum process. While the bulk of the distribution is centred around the Landauer bound, we find a few rare trajectories featuring a very large heat transfer, which are associated with non-adiabatic transitions occurring during the driving protocol. For example, consider an emission at some time, which leaves the system in its instantaneous ground state. As the eigenbasis of the Hamiltonian rotates, the state at some later time comprises a superposition of both energy eigenstates. The finite population of the excited state thus opens the possibility for a second emission to occur, potentially leading to massive overall heat transfer. An example of such a trajectory is shown in the inset of Fig. 3. On the contrary, during a classical protocol the state adiabatically follows the Hamiltonian eigenstates between jumps. This implies that an emission can only be followed by an absorption and vice versa, such that the contributions of these alternating events to the heat statistics largely cancel. We note that, apart from these rare trajectories, the heat distributions sampled from quantum protocols are very similar to their classical counterparts, with the bulk of the distribution approaching a Gaussian form as τ increases (see Appendix E, Fig. E2). The excess skewness and kurtosis of the quantum heat distributions can therefore be attributed entirely to rare, non-adiabatic processes such as the one illustrated in Fig. 3.

Even though such events are statistical outliers, they may have severe consequences for nanoscale heat management. For the data shown in Fig. 3, roughly one trajectory in every thousand involves a non-adiabatic transition. However, the maximum heat dissipated in a single trajectory is more than 30 times larger than the Landauer limit, whereas for the analogous classical protocol it is less than four times larger. This illustrates that quantum coherence drastically increases the probability of extreme heat fluctuations during the process of information erasure. Such events could damage or disrupt small-scale

quantum hardware with a low threshold of tolerance for heat dissipation. These are truly quantum fluctuations, in the sense that uncertainty in the transferred heat is increased by the existence of a coherent superposition state of the system together with the quantisation of energy exchanged with the environment. In the context of qubit erasure, these quantum fluctuations are experimentally distinguishable from thermal fluctuations since only the former involve consecutive emission or absorption events.

The results presented here can be applied to other logic operations implemented on physical hardware. Indeed, we expect that unique energetic fingerprints may also be discovered in other control protocols that process information in the quantum regime. Fast protocols that push the system far from equilibrium [59] are especially important for computing at high clock speed but are also expected to incur even greater heat fluctuations. Recently developed methods to describe dissipation in driven open quantum systems [60–63] could be used to address this problem in future work.

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[1] R. Landauer, IBM journal of research and development **5**, 183 (1961).
[2] C. H. Bennett, International Journal of Theoretical Physics **21**, 905 (1982).
[3] S. Lloyd, Nature **406**, 1047 (2000).
[4] T. Sagawa, Progress of theoretical physics **127**, 1 (2012).
[5] J. M. Parrondo, J. M. Horowitz, and T. Sagawa, Nature physics **11**, 131 (2015).
[6] J. Goold, M. Huber, A. Riera, L. del Rio, and P. Skrzypczyk, Journal of Physics A: Mathematical and Theoretical **49**, 143001 (2016).
[7] S. Vinjanampathy and J. Anders, Contemporary Physics **57**, 545 (2016).
[8] F. Binder, L. A. Correa, C. Gogolin, J. Anders, and G. Adesso, eds., *Thermodynamics in the Quantum Regime* (Springer-Verlag GmbH, 2019).
[9] M. B. Plenio and V. Vitelli, Contemporary Physics **42**, 25 (2001).
[10] K. Maruyama, F. Nori, and V. Vedral, Rev. Mod. Phys. **81**, 1 (2009).
[11] K. Sekimoto, *Stochastic energetics*, Vol. 799 (Springer, 2010).
[12] C. Jarzynski, Annu. Rev. Condens. Matter Phys. **2**, 329 (2011).
[13] U. Seifert, Rep. Prog. Phys. **75**, 126001 (2012).
[14] M. Esposito, U. Harbola, and S. Mukamel, Rev. Mod. Phys. **81**, 1665 (2009).
[15] M. Campisi, P. Hänggi, and P. Talkner, Rev. Mod. Phys. **83**, 771 (2011).
[16] P. Hänggi and P. Talkner, Nature Phys. **11**, 108 (2015).
[17] A. O. Orlov, C. S. Lent, C. C. Thorpe, G. P. Boechler, and G. L. Snider, Japanese Journal of Applied Physics **51**, 06FE10 (2012).
[18] A. Bérut, A. Arakelyan, A. Petrosyan, S. Ciliberto, R. Dillenschneider, and E. Lutz, Nature **483**, 187 (2012).
[19] Y. Jun, M. Gavrilov, and J. Bechhoefer, Phys. Rev. Lett. **113**, 190601 (2014).
[20] É. Roldán, I. A. Martinez, J. M. Parrondo, and D. Petrov, Nature Physics **10**, 457 (2014).
[21] J. Hong, B. Lambson, S. Dhuey, and J. Bokor, Science Advances **2**, e1501492 (2016).
[22] M. Gavrilov and J. Bechhoefer, Phys. Rev. Lett. **117**, 200601 (2016).
[23] J. P. Peterson, R. S. Sarthour, A. M. Souza, I. S. Oliveira, J. Goold, K. Modi, D. O. Soares-Pinto, and L. C. Céleri, Proc. Royal Soc. A **472**, 20150813 (2016).
[24] L. L. Yan, T. P. Xiong, K. Rehan, F. Zhou, D. F. Liang, L. Chen, J. Q. Zhang, W. L. Yang, Z. H. Ma, and M. Feng, Phys. Rev. Lett. **120**, 210601 (2018).
[25] R. Gaudenzi, E. Burzurí, S. Maegawa, H. van der Zant, and F. Luis, Nature Physics **14**, 565 (2018).
[26] O.-P. Saira, M. H. Matheny, R. Katti, W. Fon, G. Wimsatt, J. P. Crutchfield, S. Han, and M. L. Roukes, Phys. Rev. Research **2**, 013249 (2020).
[27] C. Jarzynski, Phys. Rev. Lett. **78**, 2690 (1997).
[28] C. Browne, A. J. P. Garner, O. C. O. Dahlsten, and V. Vedral, Phys. Rev. Lett. **113**, 100603 (2014).
[29] M. Esposito, K. Lindenberg, and C. Van den Broeck, New Journal of Physics **12**, 013013 (2010).
[30] D. Reeb and M. M. Wolf, New Journal of Physics **16**, 103011 (2014).
[31] J. Goold, U. Poschinger, and K. Modi, Phys. Rev. E **90**, 020101 (2014).
[32] J. Goold, M. Paternostro, and K. Modi, Phys. Rev. Lett. **114**, 060602 (2015).
[33] A. M. Timpanaro, J. P. Santos, and G. T. Landi, Phys. Rev. Lett. **124**, 240601 (2020).
[34] T. Albash, S. Boixo, D. A. Lidar, and P. Zanardi, New J. Phys. **14**, 123016 (2012).
[35] R. Alicki, Rep. Math. Phys. **10**, 249 (1976).
[36] P. Talkner and P. Hänggi, arXiv:1911.11660 [cond-mat.stat-mech] (2019).
[37] M. B. Plenio and P. L. Knight, Rev. Mod. Phys. **70**, 101 (1998).
[38] J. M. Horowitz, Phys. Rev. E **85**, 031110 (2012).
[39] J. M. Horowitz and T. Sagawa, J. Stat. Phys. **156**, 55 (2014).
[40] G. Manzano, J. M. Horowitz, and J. M. R. Parrondo, Phys. Rev. E **92**, 032129 (2015).
[41] G. Manzano, J. M. Horowitz, and J. M. R. Parrondo, Phys. Rev. X **8**, 31037 (2018).
[42] J. P. Pekola, P. Solinas, A. Shnirman, and D. V. Averin, New J. Phys **15**, 115006 (2013).
[43] F. Fagnola and V. Umanita, Infin. Dimens. Anal. Quantum Probab. Relat. Top. 10 **10**, 335 (2007).
[44] W. D. Roeck, C. R. Phys. **8**, 674 (2007).

[45] M. Silaev, T. T. Heikkilä, and P. Virtanen, Phys. Rev. E **90**, 022103 (2014).
[46] F. Liu, Phys. Rev. E **93**, 012127 (2016).
[47] R. Alicki, J. Phys. A **12**, L103 (1979).
[48] M. Lostaglio, D. Jennings, and T. Rudolph, Nat. Comm. **6**, 6383 (2015).
[49] A. Streltsov, G. Adesso, and M. B. Plenio, Rev. Mod. Phys. **89**, 041003 (2017).
[50] T. Baumgratz, M. Cramer, and M. B. Plenio, Phys. Rev. Lett. **113**, 140401 (2014).
[51] J. P. Santos, C. C. Lucas, G. T. Landi, and M. Paternostro, npj Quant. Info. **5**, 23 (2019).
[52] M. H. Mohammady, A. Auffèves, and J. Anders, Comm. Phys. **3**, 89 (2020).
[53] M. H. Mohammady, arXiv:arXiv:2006.07254.
[54] G. Francica, J. Goold, and F. Plastina, Phys. Rev. E **99**, 042105 (2019).
[55] M. Scandi, H. J. D. Miller, J. Anders, and M. Perarnau-Llobet, Phys. Rev. Research **2**, 023377 (2020).
[56] T. Speck and U. Seifert, Phys. Rev. E **70**, 066112 (2004).
[57] A. J. Leggett, S. Chakravarty, A. T. Dorsey, M. P. A. Fisher, A. Garg, and W. Zwerger, Rev. Mod. Phys. **59**, 1 (1987).
[58] R. Hanson, L. P. Kouwenhoven, J. R. Petta, S. Tarucha, and L. M. K. Vandersypen, Rev. Mod. Phys. **79**, 1217 (2007).
[59] K. Proesmans, J. Ehrich, and J. Bechhoefer, Phys. Rev. Lett. **125**, 100602 (2020).
[60] R. Dann, A. Tobalina, and R. Kosloff, Phys. Rev. Lett. **122**, 250402 (2019).
[61] R. Dann, A. Tobalina, and R. Kosloff, Phys. Rev. A **101**, 052102 (2020).
[62] N. Pancotti, M. Scandi, M. T. Mitchison, and M. Perarnau-Llobet, Phys. Rev. X **10**, 031015 (2020).
[63] M. Popovic, M. T. Mitchison, A. Strathearn, B. W. Lovett, J. Goold, and P. R. Eastham, arXiv:2008.06491 [quant-ph] .
[64] H. J. D. Miller, M. H. Mohammady, M. Perarnau-Llobet, and G. Guarnieri, (2020), arXiv:2006.07316 [quant-ph].
[65] K. Brandner, M. Bauer, and U. Seifert, Phys. Rev. Lett. **119**, 170602 (2017).
[66] V. Cavina, A. Mari, and V. Giovannetti, Phys. Rev. Lett. **119**, 050601 (2017).
[67] M. Scandi and M. Perarnau-Llobet, Quantum **3**, 197 (2019).
[68] H. J. D. Miller, M. Scandi, J. Anders, and M. Perarnau-Llobet, Phys. Rev. Lett. **123**, 230603 (2019).
[69] D. Petz and F. Hiai, *Introduction to Matrix Analysis and Applications* (Springer-Verlag, New Delhi, 2014).
[70] Á. M. Alhambra and M. P. Woods, Phys. Rev. A **96**, 1 (2017).
[71] M. Lostaglio, K. Korzekwa, D. Jennings, and T. Rudolph, Phys. Rev. X **5**, 021001 (2015).
[72] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes* (Cambridge University Press, Cambridge, 2007).

A. Proof of (5)

In this Appendix, we will show how to decompose the CGF for excess heat into a quantum and classical contribution. As stated in the main text, we assume that the quantum Markovian semigroup governing the open system dynamics satisfies the detailed balance condition $\mathscr{L}_t^*(.) = \tilde{\mathscr{L}}_t(.) + 2i[\hat{H}_t, (.)]$, where $\tilde{\mathscr{L}}_t$ is a dual generator determined by $\mathrm{Tr}\left(\tilde{\mathscr{L}}_t(\hat{A})\hat{\pi}_t\hat{B}\right) = \mathrm{Tr}\left(\hat{A}\hat{\pi}_t\mathscr{L}_t^*(\hat{B})\right)$ for any bounded operators $\hat{A}, \hat{B}$ [35]. Here, and throughout this Supplemental Material, we set $\hbar = 1$. We also assume $\hat{H}_t$ is non-degenerate for simplicity. Note that while this condition remains valid at the beginning of the protocol, the difference in energy eigenvalues are negligible when compared to the temperature of the environment. It has been proven that this condition ensures that $\tilde{\mathscr{L}}$ also forms a quantum Markovian semigroup (Lemma 5.1, Ref. [43]), which in turn implies that the generator satisfies the following time-translational symmetry (Theorem 3.1, Ref. [43]):

$$\mathscr{L}_t^*(.) = \hat{\pi}_t^{i\alpha}\mathscr{L}_t^*\left(\hat{\pi}_t^{-i\alpha}(.)\hat{\pi}_t^{i\alpha}\right)\hat{\pi}_t^{-i\alpha}, \qquad \forall\alpha \in \mathbb{R}. \tag{A1}$$

Since the fixed point is thermal, and hence a function of the Hamiltonian, we may write this condition in the equivalent form

$$[\mathscr{L}_t^*, \mathscr{H}_t] = 0, \tag{A2}$$

where $\mathscr{H}_t(.) = [\hat{H}_t, (.)]$. A useful identity for the dephasing operation $\mathcal{D}_{\hat{H}_t}(.) = \sum_n |n_t\rangle\langle n_t|(.)|n_t\rangle\langle n_t|$ is via the infinite time-average for non-degenerate Hamiltonians, namely

$$\mathcal{D}_{\hat{H}_t}(.) = \lim_{\mathcal{T}\to\infty}\frac{1}{\mathcal{T}}\int_0^{\mathcal{T}} d\mu\; e^{-i\mu\mathscr{H}_t}(.), \tag{A3}$$

Using (A2) this implies commutation between the adjoint Lindblad evolution (at fixed t) and the dephasing operation, i.e.

$$[\mathcal{D}_{\hat{H}_t}, e^{\nu\mathscr{L}_t^*}] = [\mathcal{D}_{\hat{H}_t}, e^{\nu\mathscr{L}_t}] = 0, \tag{A4}$$